4 **(a)** The first four terms of a sequence are 2, 3, 0, 3 and the subsequent terms are given by $a_{k+4} = a_k$.

(i) State what type of sequence this is. **[1]**

(ii) Find $\sum_{k=1}^{200} a_k$. **[1]**

(b) A different sequence is given by $u_n = b^n$ where b is a constant and $n \geqslant 1$.

(i) State the set of values of b for which this is a divergent sequence. **[2]**

(ii) In the case where $b = \frac{1}{3}$, find the sum of all the terms in the sequence. **[2]**